

Order Relations on a Set

Directed Set: (1), (2)

Partially Ordered Set: (1), (2), (3)

Totally Ordered Set: (1), (2), (3), (4)

1. Reflexive: $(\forall a \in A) a \leq a$
2. Transitive: $(\forall a, b, c) a \leq b \text{ and } b \leq c \Rightarrow a \leq c$
3. Antisymmetric: $(\forall a, b) a \leq b \text{ and } b \leq a \Rightarrow a = b$
4. Connex: $(\forall a, b) a \leq b \text{ or } b \leq a$